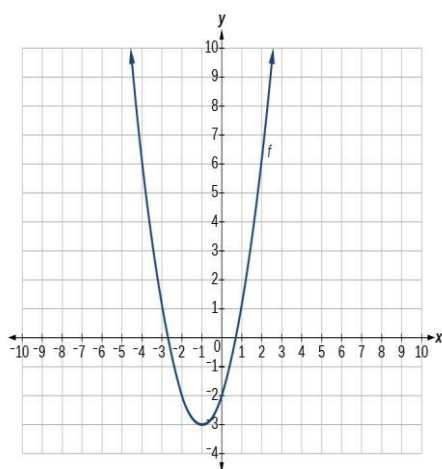
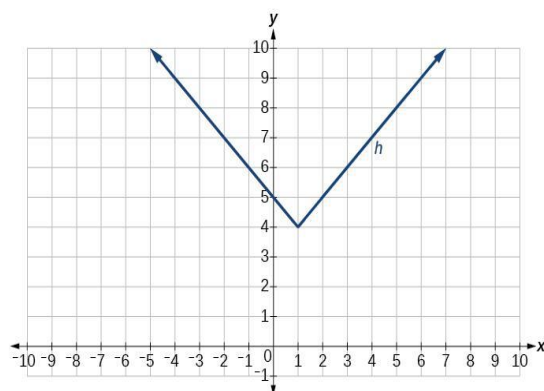


## Section 1.5 Exercise Answers

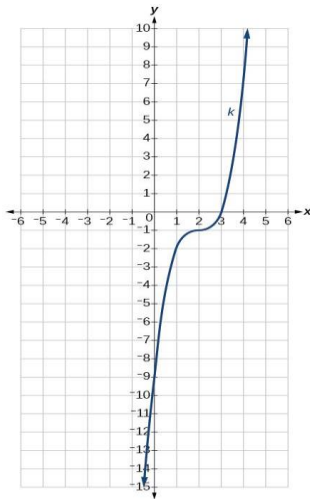
1.  $g(x) = \sqrt{x+2} + 1$
2.  $g(x) = |x-1| - 3$
3.  $g(x) = \frac{1}{x-3} - 4$
4.  $g(x) = \frac{1}{(x+4)^2} + 2$
5. Shifted right 49 units
6. Shifted left 43 units
7. Shifted left 3 units
8. Shifted right 4 units
9. Shifted up 5 units
10. Shifted up 8 units
11. Shifted down 2 units
12. Shifted down 7 units
13. Shifted right 2 units and up 3 units.
14. Shifted left 4 units down 1 unit
15. Shifted up 1 unit
16. Shifted down 3 units
17. Shifted right 1 unit
18. This is a parabola that has been shifted down 3 units and to the left 1 unit.



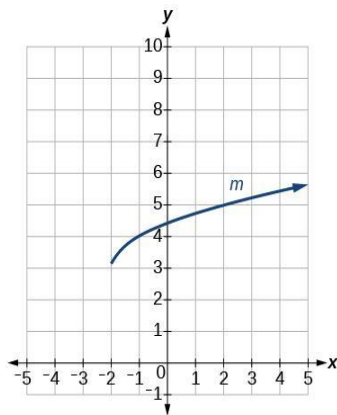
19. This is an absolute value function that has been shifted up 4 units and to the right 1 unit.



20. This is a cubic function that has been shifted down 1 unit and to the right 2 units.



21. This is a square root function that has been shifted up 3 units and to the left 2 units.



22.  $f(x) = |x - 3| - 2$

23.  $f(x) = (x - 1)^2 - 3$

24.  $f(x) = \sqrt{x + 3} - 1$

25.  $f(x) = |x + 2| + 2$

26.  $f(x) = (x - 2)^2$

27.  $f(x) = \sqrt{x + 3}$

28.  $f(x) = |x + 3| - 2$

29. Even Function

30. Neither

31. Odd Function

32. Neither

33. Even Function

34. Odd Function

35. Reflection across the  $x$ -axis

36. Reflection across the  $y$ -axis

- 37. Vertical stretch by a factor of 4.
- 38. Vertical stretch by a factor of 6
- 39. Horizontal shrink by a factor of 5
- 40. Horizontal shrink by a factor of 2
- 41. Horizontal stretch by a factor of 3
- 42. Horizontal stretch by a factor of 5
- 43. Vertical stretch by a factor of 3 and a reflection across the  $y$ -axis
- 44. Reflection across the  $x$ -axis and a horizontal shrink by a factor of 3